# 2026 SE&ATN Symposium Abstract Selection

Submit Abstract

Submission Deadline
May 19, 2026

## Review Your Submission

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## Submission Details

RTX Business Unit
Raytheon

Submission Title
AI-Driven Bayesian Templates for Hubbard's AIE in engineering decisions under uncertainty

Session Type
Open Presentation (International Friendly, no ITAR/EAR content)

Presentation Type
Tutorial

Track
RTX Tech Roadmap - Artificial Intelligence

Submission Details
When facing engineering decisions under uncertainty—sparse failure data, variable costs, etc - Doug Hubbard's Applied Information Economics (AIE) offers a framework that integrates calibrated expert estimates, Bayesian updating, Monte Carlo simulation, and Value of Information (VoI) analysis. However, two barriers block widespread adoption: (1) creating quality probability distributions, and (2)

determining which pieces of information matter and where they fit into Bayesian models (as priors, likelihoods, evidence, or posteriors).

The tutorial removes barriers using workflow templates (Excel for accessibility, Jupyter Notebooks for scalability) that work with an AI assistant, taking a problem description plus sample data and breaking the problem into modular "measurement blocks," each specifying:
- Key uncertainties and decision variables, Appropriate prior distributions (seeded by calibrated estimates or fitted from data), Likelihood models for new evidence, Exact placement in the Bayesian chain or network, Monte Carlo parameters for propagation and VoI calculations. Generated with step explanations: why specific data belongs in a block, how the Bayesian update formula applies, the rationale for the chosen distribution (e.g., Beta for binomial reliability, Weibull for time-to-failure), and how to reduce bias through calibrated estimation.

Tutorial Format (Hands-On):
10-20 min – Concise overview of AIE principles with relevant R&M examples (failure rate updating, maintenance optimization, risk ranking).
20-40 min – Live demonstration of effective prompting techniques that produce high-quality JSON outputs, followed by end-to-end template workflow.
20-40 min – Guided participant exercises on real R&M scenarios (e.g., updating component reliability with limited tests; deciding on additional inspections via Expected Value of Perfect Information).
10-20 min – Open discussion on validation, scaling to complex systems, common pitfalls, and extensions (digital thread integration, batch processing).

Attendees receive the template package and step guides to apply the method to their own problems. The tutorial aligns with risk analysis, AI/ML applications, and decision support under uncertainty demonstrating how AI-augmented templates overcome long-standing implementation challenges, enabling robust Bayesian solutions from problem descriptions and data without getting lost in the details. No advanced statistics or programing background required.

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